Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Firewall Basics

Firewall types and use cases (hardware and software firewalls). Configure basic rules in a software firewall (e.g., Windows Defender).

Task: Document how to block and allow specific traffic using a firewall.

❖ 1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

➢ Aim

- To understand the basics and types of firewalls and configure a new outbound rule in Windows Defender Firewall to block all outgoing network traffic from the Spotify desktop application.

➢ Introduction

- A **firewall** is a network security system that monitors and controls incoming and outgoing network traffic according to predefined security rules. It helps protect a computer or network from unauthorized access and unwanted communication.
- Firewalls can allow or block traffic based on factors such as application, IP address, port number, protocol, and network profile.

- For this practical, Windows Defender Firewall is used to create an outbound rule that blocks network communication from the Spotify desktop application.

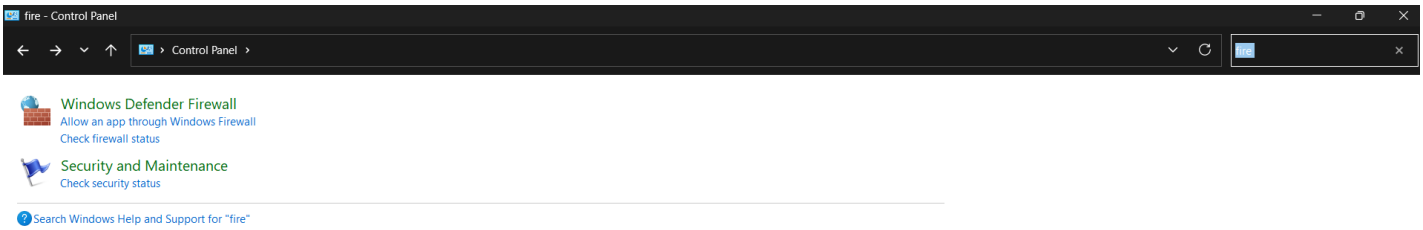
➢ **Types of Firewalls**
➢ **1. Hardware Firewall**
➢ A hardware firewall is a physical security device that monitors and filters network traffic between networks.
➢ **Example:** A firewall integrated into a router or a dedicated network security appliance.
➢ **Use case:** Protecting an office network by controlling traffic entering or leaving the organization.

➢ **2. Software Firewall**
➢ A software firewall is a program or operating-system feature that controls network traffic on an individual computer.
➢ **Example:** Windows Defender Firewall.
➢ **Use case:** Blocking a particular application from accessing the Internet or allowing only approved applications to communicate over a network.

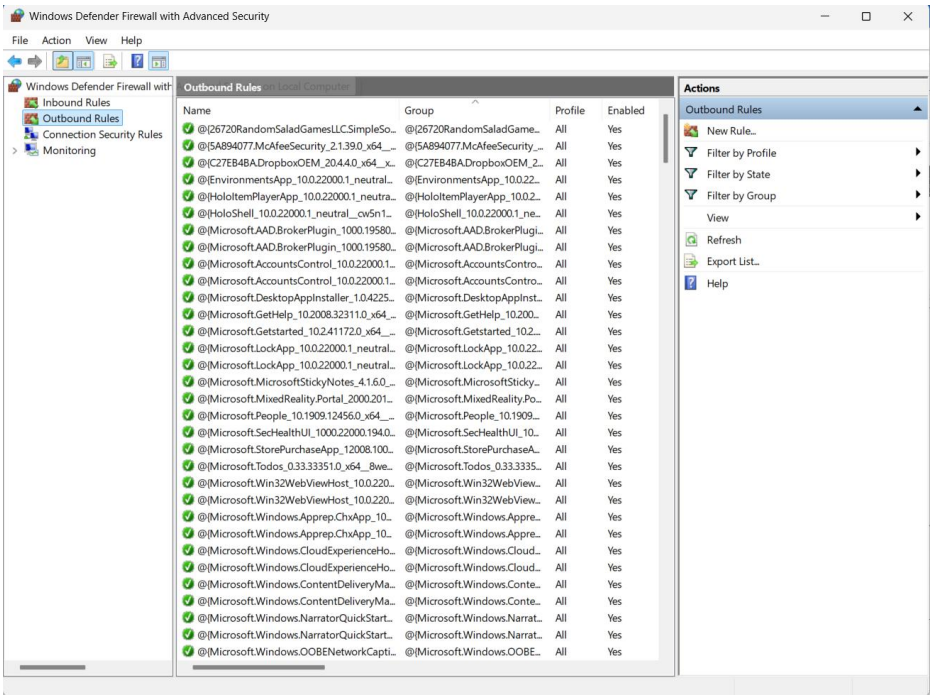
➢ **Tools and Requirements**
- A Windows computer or laptop
- Windows Defender Firewall
- Spotify desktop application
- Administrator permission to create firewall rules

➢ **Procedure: Creating an Outbound Rule to Block Spotify**

- **Step 1: Open Windows Defender Firewall**
  1. Open the **Start** menu.
  2. Search for **Windows Defender Firewall with Advanced Security**.
  3. Open the application. Administrator permission may be required to create a rule.

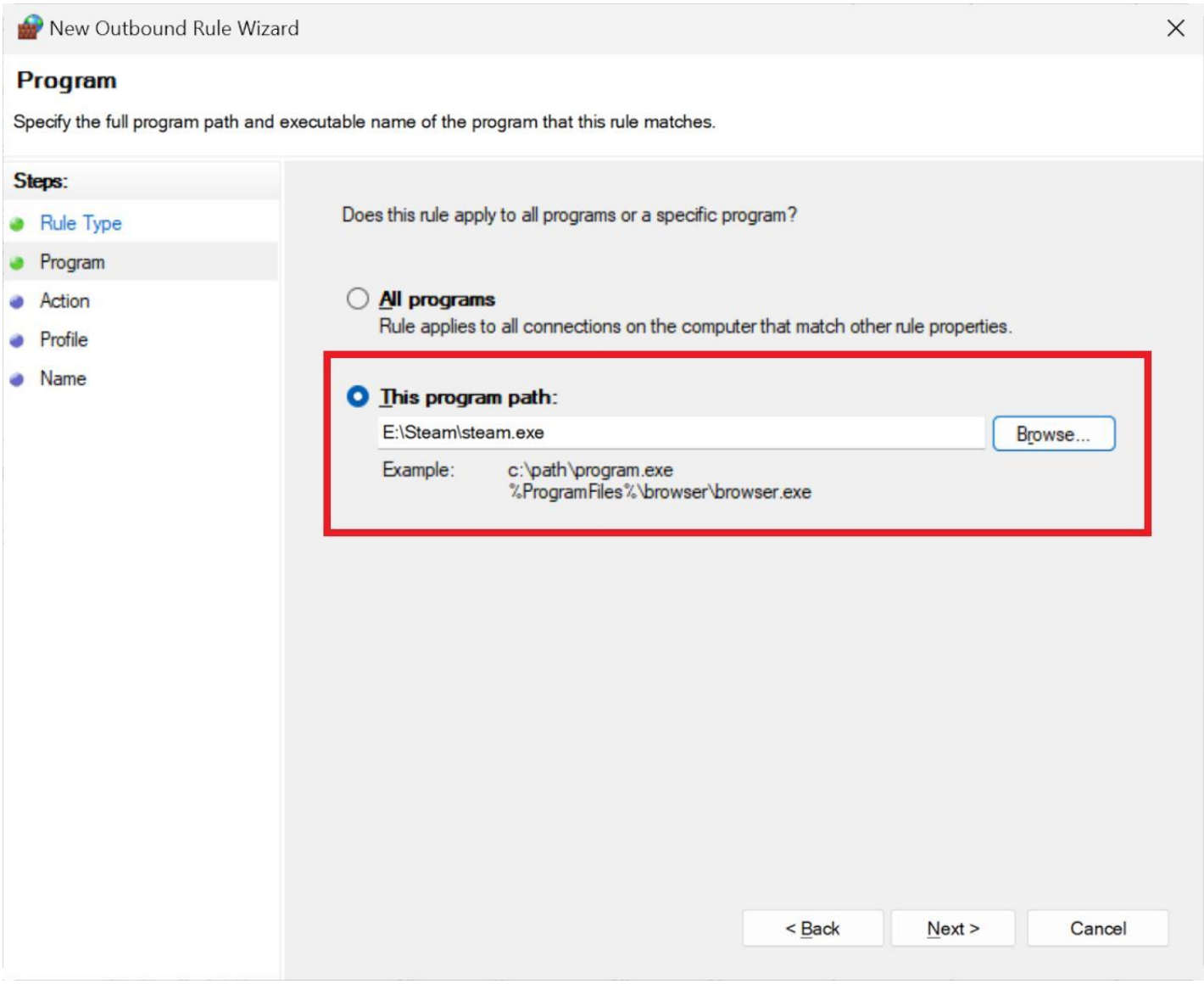


- **Step 2: Open Outbound Rules**
  1. In the left-side panel, click **Outbound Rules**.
  2. In the right-side panel, click **New Rule...**.
  3. The New Outbound Rule Wizard will open.



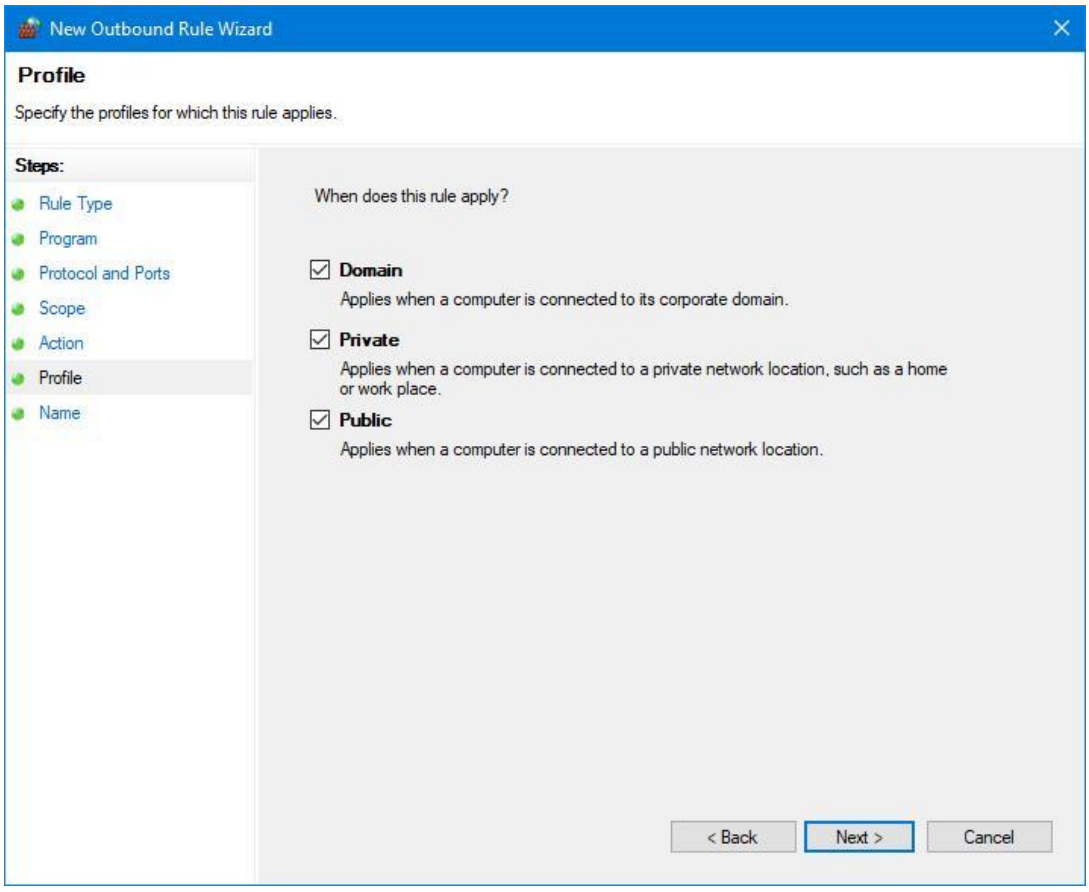
- **Step 3: Select the Rule Type**
  1. Select **Program**.
  2. Click **Next**.
  3. This option allows the firewall rule to apply to a particular application.

- **Step 4: Select the Spotify Program**
  1. Select **This program path**.
  2. Browse to the Spotify desktop application's executable file, usually named Spotify.exe.
  3. Select the correct executable file and click **Next**.
- The exact file location may vary depending on how Spotify was installed. Select the executable for the desktop application rather than a web browser.



- **Step 5: Block the Connection**
  1. Select **Block the connection**.
  2. Click **Next**.
- This setting instructs Windows Defender Firewall to block outgoing network connections matching this rule.

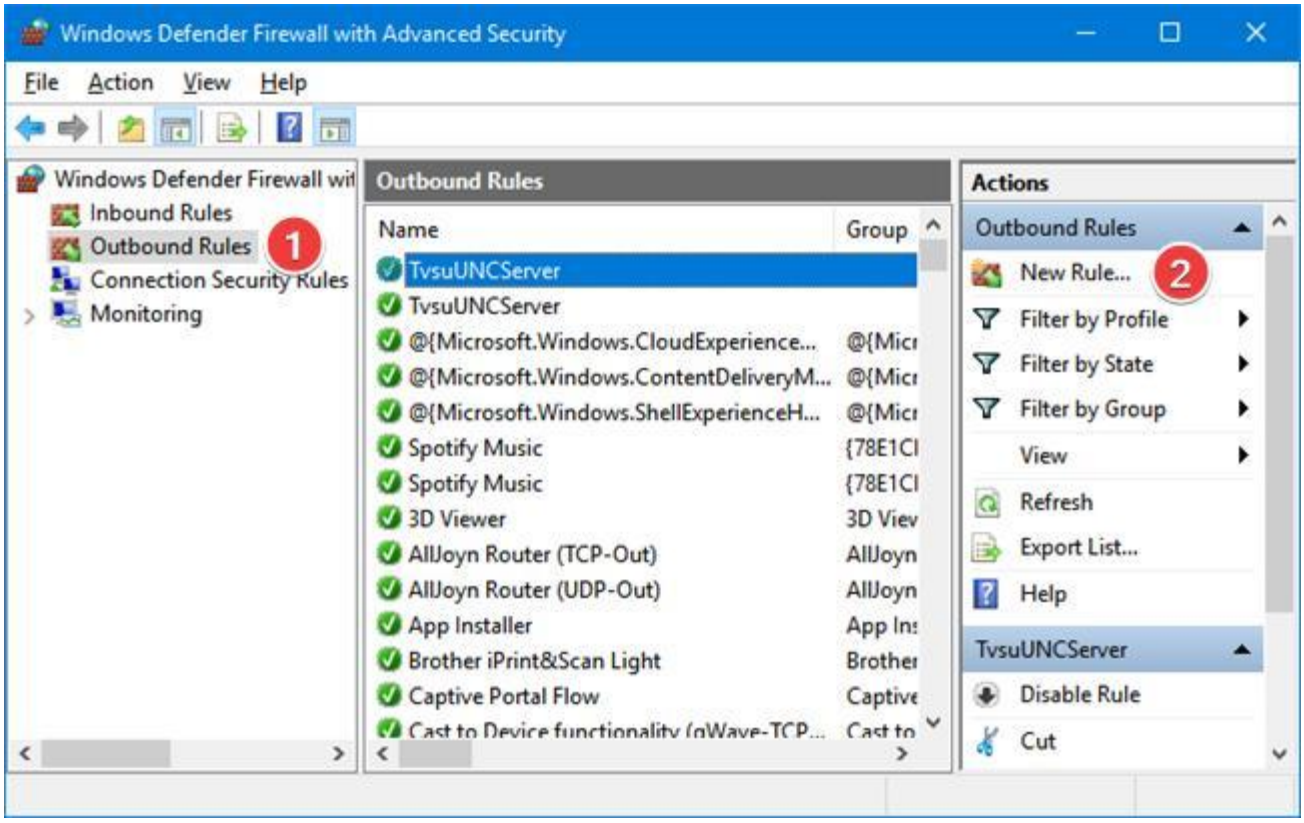
- **Step 6: Select Network Profiles**
- Select the profiles on which the rule should apply:
  - **Domain:** For computers connected to a domain network.
  - **Private:** For trusted networks, such as a home network.
  - **Public:** For public networks, such as public Wi-Fi.
- For this assignment, select all three profiles if the intention is to block Spotify on every supported profile. On a managed computer, follow the administrator's instructions.
- Click **Next**.



➢ **Step 7: Name the Rule**

1. Enter the rule name: Block Spotify Outbound Traffic.
2. In the description field, enter: Blocks outgoing network connections from the Spotify desktop application.
3. Click **Finish**.

➢ The new rule should now appear in the Outbound Rules list.



➢ **Rule Configuration Summary**

Setting	Configuration
Rule Type	Program

Program	Spotify desktop executable (Spotify.exe)
Direction	Outbound
Action	Block the connection
Profiles	Domain, Private, and Public
Rule Name	Block Spotify Outbound Traffic

- **How the Firewall Rule Works**
- When Spotify attempts to establish an outgoing network connection, Windows Defender Firewall checks the applicable rules. If the connection matches the enabled blocking rule, the firewall blocks that traffic.
- As a result, Spotify may be unable to stream music, load online content, or access other online services. Other applications should remain unaffected by this program-specific rule, unless another rule or security control also affects them.

- **Verification**
  1. Open **Windows Defender Firewall with Advanced Security**.
  2. Select **Outbound Rules**.
  3. Find Block Spotify Outbound Traffic.
  4. Confirm that the rule is enabled and its action is set to **Block**.
  5. Open the Spotify desktop application and check whether it can access online content.
- If the rule is configured correctly and matches Spotify's traffic, the application should be unable to access the blocked network services. If it

continues to work, check that the correct executable path was selected and that the rule is enabled.

- **Result**
- A new outbound firewall rule was configured in Windows Defender Firewall to block network traffic from the Spotify desktop application. The rule can be verified in the Outbound Rules section.

❖ 2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

- **Aim**
  - To configure an inbound rule in Windows Defender Firewall to allow incoming connections for Google Chrome on port 443 (HTTPS).

- **Introduction**
  - A **firewall** is a security system that monitors and controls incoming and outgoing network traffic according to predefined rules. Windows Defender Firewall is a software firewall available in Microsoft Windows that helps protect a computer from unauthorized network access.
  - An **inbound rule** controls network traffic coming into a computer. In this practical, an inbound rule is created for Google Chrome using TCP port 443, which is commonly used for HTTPS communication.

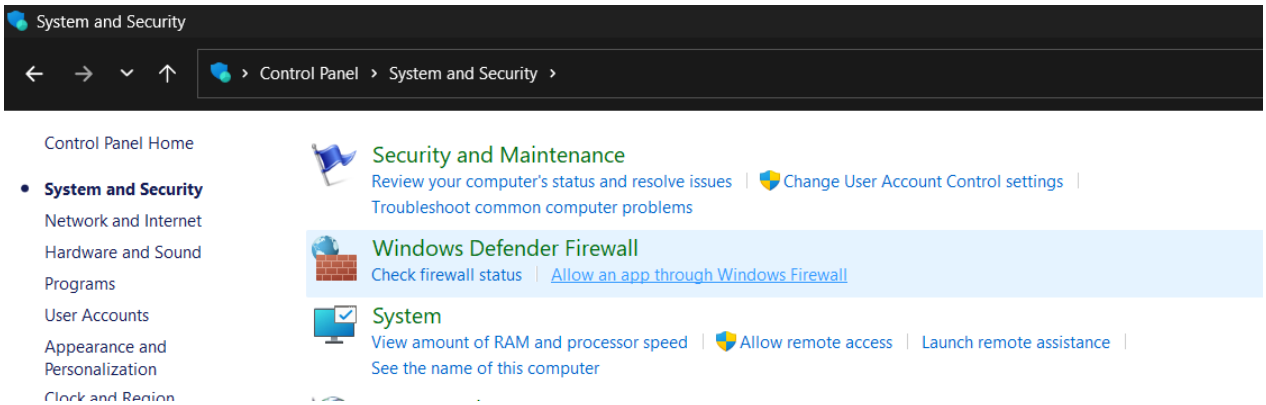
- **Tools and Requirements**
  - A Windows computer or laptop

- Windows Defender Firewall with Advanced Security
- Google Chrome installed
- Administrator permission to create firewall rules

➢ Procedure

➢ Step 1: Open Windows Defender Firewall

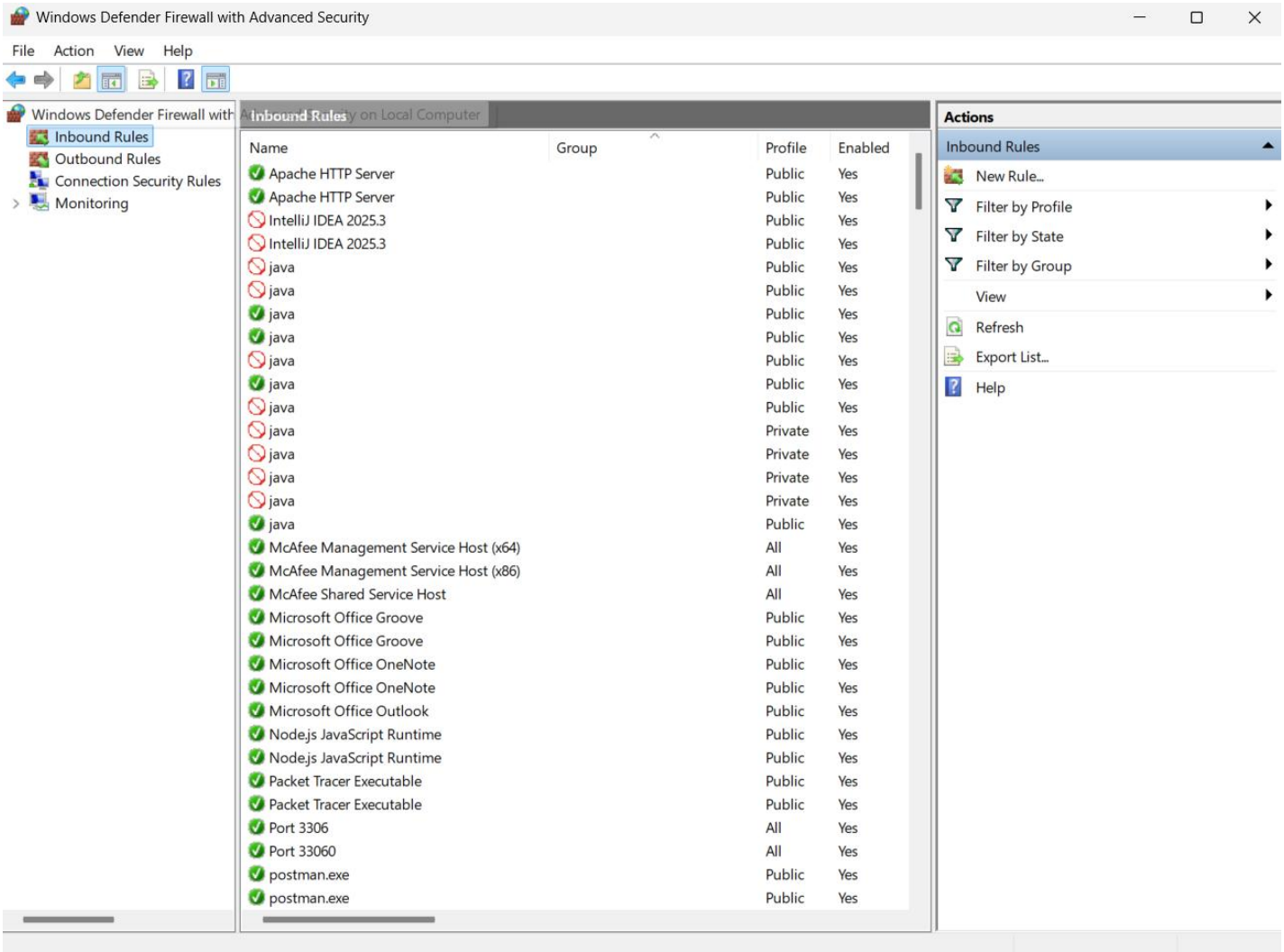
1. Open the **Start** menu.
2. Search for **Windows Defender Firewall with Advanced Security**.
3. Open the application. Administrator permission may be required.



➢ Step 2: Open Inbound Rules

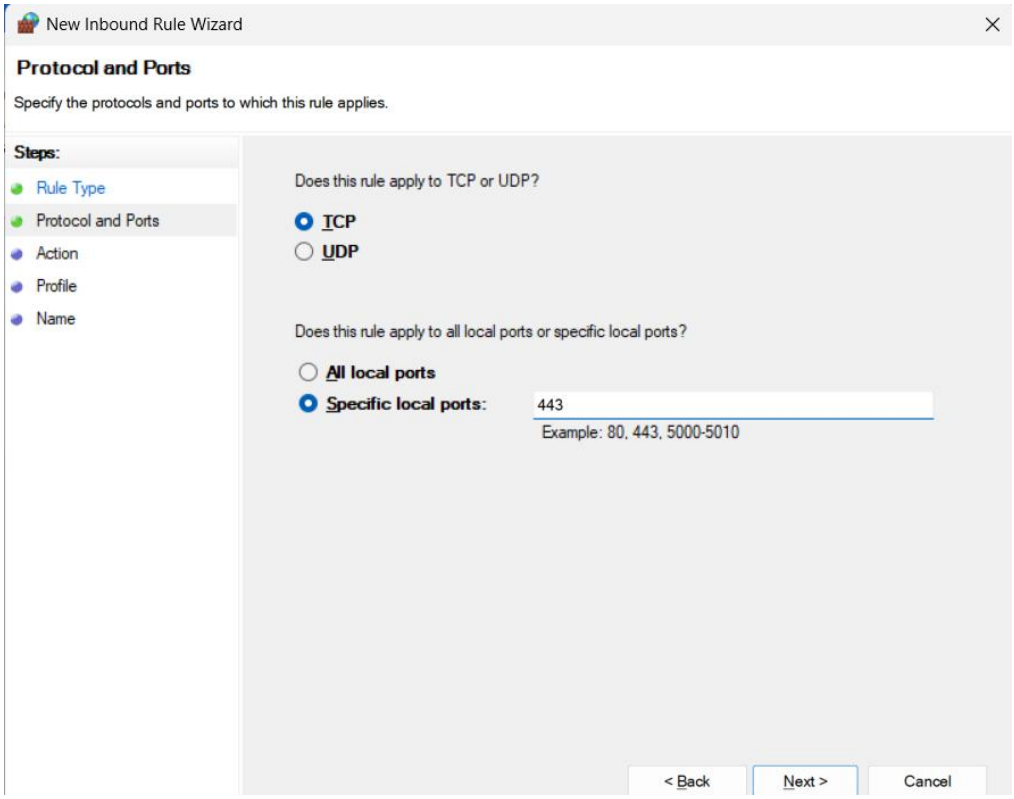
1. In the left-side panel, click **Inbound Rules**.
2. In the right-side panel, click **New Rule...**.

3. The New Inbound Rule Wizard will open.



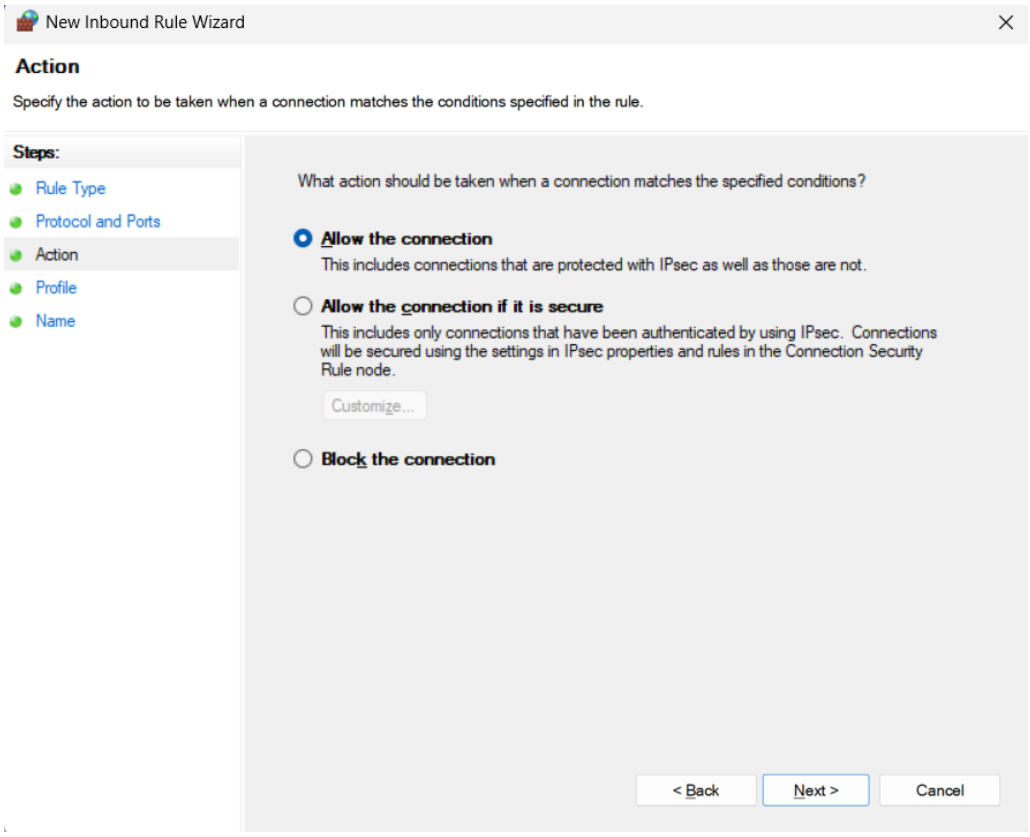
- **Step 3: Select the Rule Type**
  1. Select **Port**.
  2. Click **Next**.

- This option allows the firewall rule to be configured using a specific network port and protocol.
- **Step 4: Select TCP and Port 443**
  1. Select **TCP**.
  2. Under **Specific local ports**, enter 443.
  3. Click **Next**.
- TCP port 443 is commonly used for HTTPS communication.



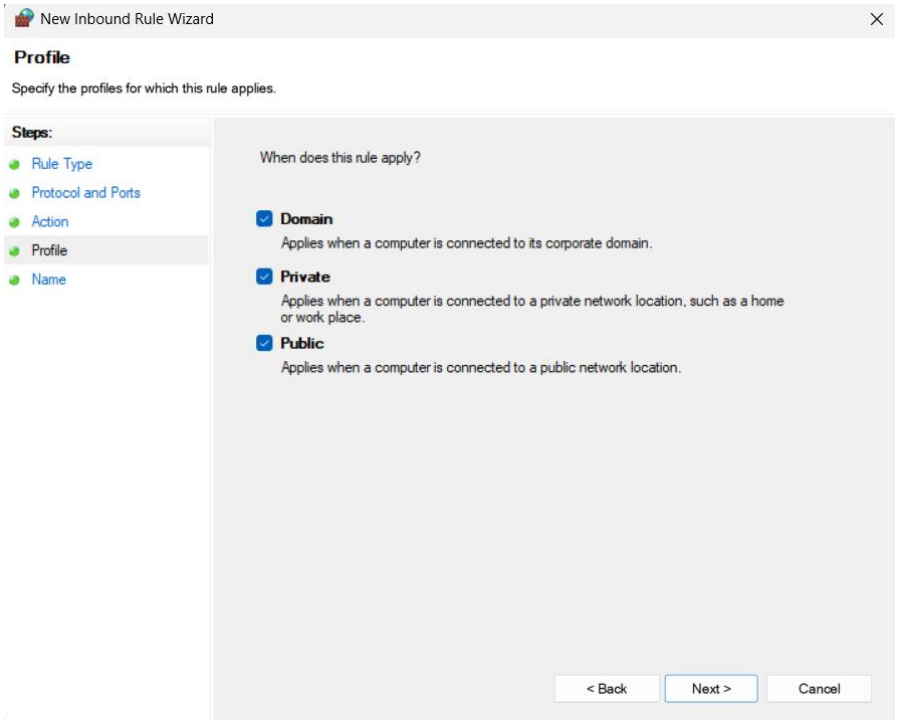
- **Step 5: Allow the Connection**
  1. Select **Allow the connection**.
  2. Click **Next**.

- This action permits incoming connections that match the rule's conditions, subject to other applicable firewall and network controls.

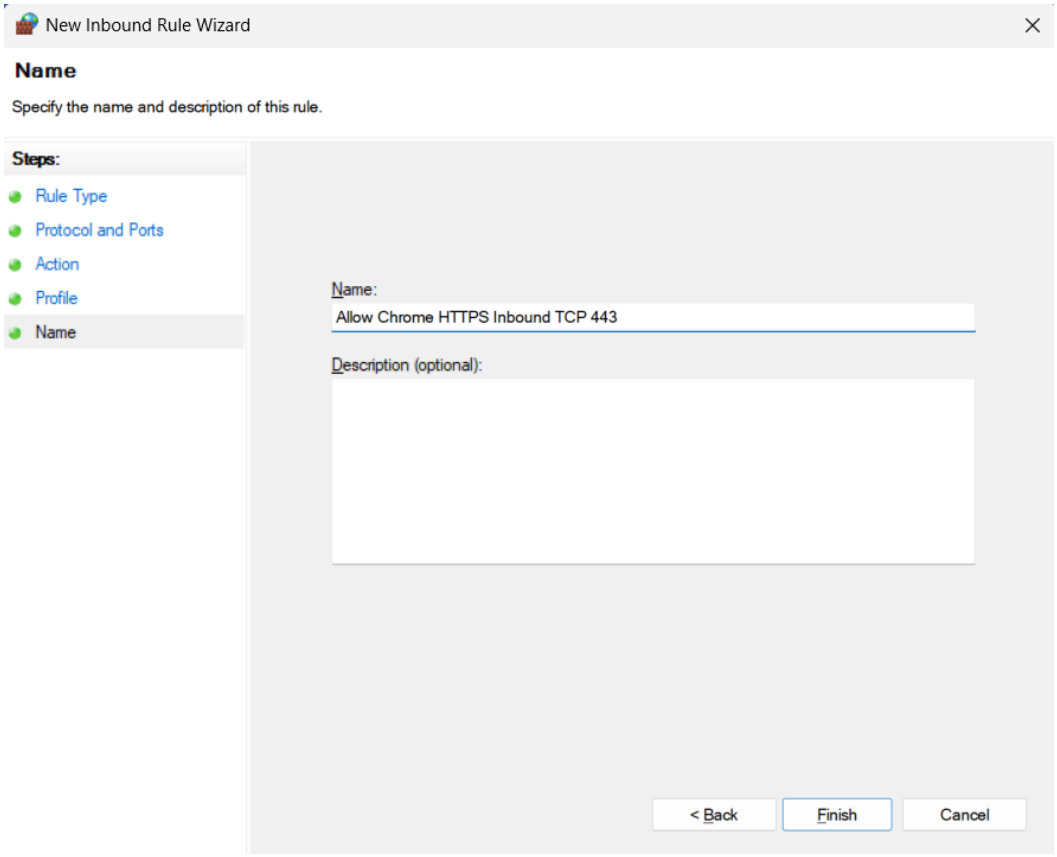


- **Step 6: Select Network Profiles**
- Select the profiles on which the rule should apply:
  - **Domain:** For computers connected to an organizational domain.
  - **Private:** For trusted networks, such as a home network.
  - **Public:** For public networks, such as public Wi-Fi.
- For a practical demonstration, select the profile required by your instructor. Select all three only if the rule is intended to apply to all profiles.

- Click **Next**.

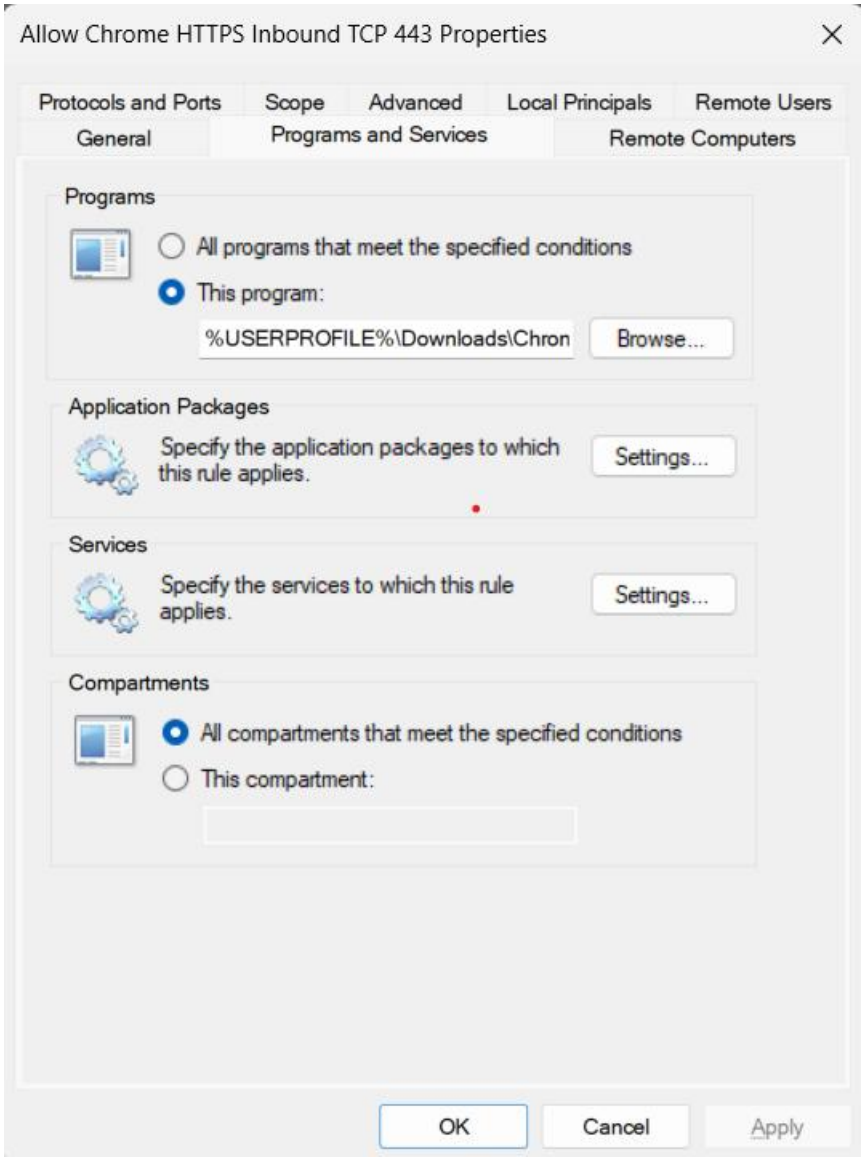


- **Step 7: Name the Rule**
- Enter the following details:
  - **Name:** Allow Chrome HTTPS Inbound TCP 443
  - **Description:** Allows incoming TCP connections on local port 443 for the specified Chrome program.
- Click **Finish**.



- **Step 8: Restrict the Rule to Google Chrome**
- A port-based rule alone applies to matching incoming traffic on port 443; it does not restrict traffic to Chrome.
- To restrict the rule to Google Chrome:
  1. Locate the newly created rule in **Inbound Rules**.
  2. Right-click it and select **Properties**.
  3. Open the **Programs and Services** tab, if available.
  4. Under **Programs**, select **This program** and browse to the actual Google Chrome executable, usually named chrome.exe.
  5. Apply the changes.

- If the Windows version does not expose this setting for the selected rule, create a **Custom** inbound rule instead, specify the Chrome executable under Program, select TCP and local port 443, and choose **Allow the connection**. Complete the remaining wizard steps and give the rule a descriptive name.



- **Rule Configuration Summary**

Setting	Configuration
Direction	Inbound
Rule Type	Port, or Custom when restricting by program
Program	Google Chrome executable (chrome.exe)
Protocol	TCP
Local Port	443
Action	Allow the connection
Profiles	Selected according to the practical requirements
Rule Name	Allow Chrome HTTPS Inbound TCP 443

- **Verification**
  1. Open **Windows Defender Firewall with Advanced Security**.
  2. Select **Inbound Rules**.
  3. Locate Allow Chrome HTTPS Inbound TCP 443.
  4. Confirm that the rule is enabled and its action is **Allow**.
  5. Open the rule's properties and verify the program, protocol, port, and selected profiles.
- The rule's presence confirms that it has been configured. It does not, by itself, prove that an incoming connection can reach Chrome; successful

communication also depends on the network path, listening service, and other security settings.

- **Result**
- An inbound firewall rule was configured to allow matching incoming TCP connections on local port 443 for the specified Google Chrome executable. The rule's settings can be checked in the Inbound Rules section of Windows Defender Firewall.

❖ **3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Use the 'Scope' tab when creating a new rule to specify the IP address.**

- **Aim**
  - To create a firewall rule in Windows Defender Firewall to block network traffic from a specific IP address, 123.45.67.89, using the Scope tab.

- **Introduction**
  - Windows Defender Firewall is a software firewall that monitors and controls incoming and outgoing network traffic according to configured security rules.
  - The **Scope tab** allows an administrator to specify the local and remote IP addresses to which a firewall rule applies. By defining a specific remote IP address and selecting the **Block the connection** action, traffic matching that address can be blocked.

- In this practical, the example IP address 123.45.67.89 is used to demonstrate how to create a blocking rule.

➢ **3. Tools and Requirements**

- A Windows computer or laptop
- Windows Defender Firewall with Advanced Security
- Administrator permission to create firewall rules
- Example IP address: 123.45.67.89

➢ **Procedure**

➢ **Step 1: Open Windows Defender Firewall**

1. Open the **Start** menu.
2. Search for **Windows Defender Firewall with Advanced Security**.
3. Open the application.

➢ **Step 2: Create a New Inbound Rule**

1. Select **Inbound Rules** from the left-side panel.
2. Click **New Rule...** from the right-side panel.
3. Select **Custom** as the rule type.
4. Click **Next**.

➢ A custom rule allows the administrator to configure additional conditions, including IP address scope.

➢ **Step 3: Specify the Program**

1. Select **All programs** if the intention is to block matching traffic regardless of the application.
2. Click **Next**.

➢ This makes the rule independent of a particular application.

➢ **Step 4: Configure Protocol and Ports**

1. On the **Protocol and Ports** page, select **Any** under Protocol type if all IP traffic from the specified address should be blocked.
2. Leave the port settings unrestricted as appropriate for the selected protocol.
3. Click **Next**.

- This ensures that the rule is not limited to a single TCP or UDP port.
- **Step 5: Configure the Scope Tab**
  1. On the **Scope** page, locate **Which remote IP addresses does this rule apply to?**
  2. Select **These IP addresses**.
  3. Click **Add**.
  4. Enter the example IP address 123.45.67.89.
  5. Click **OK** to add the address.
  6. Leave the local IP address scope as **Any IP address**, unless a specific local address is required.
  7. Click **Next**.
- The remote IP address identifies the source of the incoming traffic that the rule will match.
- **Step 6: Block the Connection**
  1. Select **Block the connection**.
  2. Click **Next**.
- This instructs Windows Defender Firewall to block connections matching the configured rule.
- **Step 7: Select Network Profiles**
- Select the profiles where the rule should apply:
  - **Domain:** For computers connected to an organizational domain.
  - **Private:** For trusted networks, such as a home network.

- **Public:** For public networks.

➢ For this assignment, select all three profiles if the rule should apply across all of them. Otherwise, choose the profiles required by the practical.
➢ Click **Next**.
➢ **Step 8: Name the Rule**
➢ Enter the following details:
- **Name:** Block IP 123.45.67.89
- **Description:** Blocks incoming network traffic from the specified remote IP address.

➢ Click **Finish**.
➢ The rule should now appear in the Inbound Rules list.
➢ **Rule Configuration Summary**

Setting	Configuration
Rule Type	Custom
Program	All programs
Protocol	Any
Direction	Inbound
Remote IP Address	123.45.67.89
Local IP Address	Any IP address
Action	Block the connection
Profiles	Domain, Private, and Public, if required

Rule Name	Block IP 123.45.67.89

- **Verification**
  1. Open **Windows Defender Firewall with Advanced Security**.
  2. Select **Inbound Rules**.
  3. Find Block IP 123.45.67.89.
  4. Confirm that the rule is enabled.
  5. Open its properties and select the **Scope** tab.
  6. Verify that 123.45.67.89 appears under remote IP addresses.
  7. Confirm that the action is set to **Block the connection**.
- The rule's settings can be verified in the firewall console. Actual blocking depends on whether traffic from that address reaches the computer and matches the rule.

- **Result**
- A custom inbound firewall rule was configured to block matching incoming IP traffic from the example remote address 123.45.67.89. The remote address was specified in the Scope tab, and the rule can be verified in the Inbound Rules section.

❖ 4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

➢ Aim

- To compare hardware firewalls and software firewalls by identifying three key differences and explaining one practical use case for each.

➢ Introduction

- A **firewall** is a security system that monitors and controls incoming and outgoing network traffic according to predefined security rules. It helps protect computers and networks from unauthorized access and unwanted communication.
- Firewalls are mainly classified into two types: **hardware firewalls** and **software firewalls**. Both improve network security, but they differ in their implementation, protection scope, and management.

➢ Three Key Differences

Sr. No.	Hardware Firewall	Software Firewall
1. Implementation	A physical network device or security appliance that filters network traffic.	A program or operating-system feature installed on an individual computer or server.

2. Protection Scope	Can protect multiple devices across a network when deployed at an appropriate network boundary.	Primarily protects the individual computer or server on which it operates.
3. Management and Use	Usually configured centrally by a network administrator to manage network-wide security policies.	Can be configured on each computer to control application-specific or device-specific traffic.

- **Use Case of a Hardware Firewall**
- **Example: Large Office Network**
  - A large office may use a hardware firewall between its internal network and the Internet. The firewall can monitor traffic entering and leaving the office, enforce security policies, and help prevent unauthorized access to internal systems.
  - **Use case:** Protecting multiple office computers, servers, and other network devices through centrally managed network security rules.

- **Use Case of a Software Firewall**
- **Example: Windows Defender Firewall**
  - Windows Defender Firewall is a software firewall built into Microsoft Windows. It allows users and administrators to create rules that control incoming and outgoing traffic on an individual computer.

- **Use case:** Blocking a particular application from accessing the Internet or allowing only specified network connections on a laptop or desktop computer.

➢ **Result**

➢ Hardware and software firewalls were compared based on their implementation, protection scope, and management. A hardware firewall is useful for protecting an entire network, while a software firewall provides traffic control on an individual device.